

Create a working custom VPC. Launch one server in each subnet & connect private server with bastion host using CLI.

Prerequisites

- AWS CLI
- AWS Account

Configure AWS CLI: **aws configure**

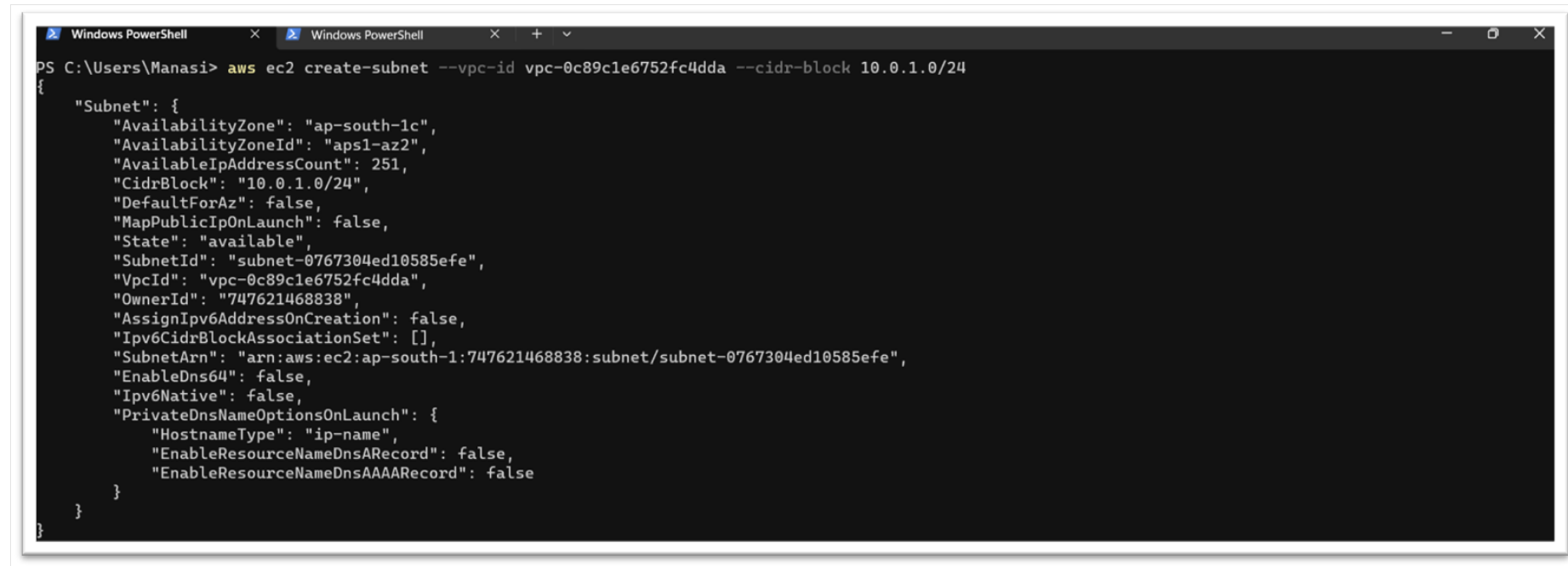
Create VPC: **aws ec2 create-vpc --cidr-block 10.0.0.0/16**

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

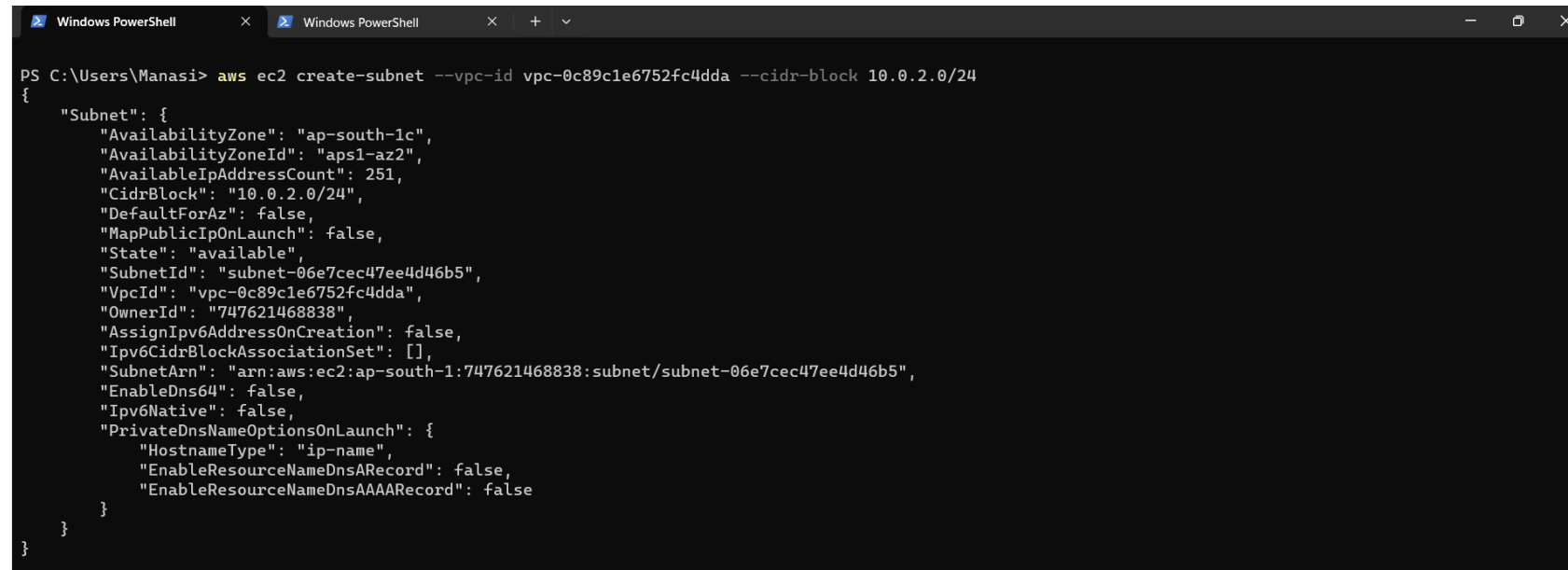
PS C:\Users\Manasi> aws configure
AWS Access Key ID [*****AY73]: AKIA24ENWJKTJKXPAY73
AWS Secret Access Key [*****1t6D]: 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
Default region name [ap-south-1]: ap-south-1
Default output format [json]: json
PS C:\Users\Manasi> #create Vpc
PS C:\Users\Manasi> aws ec2 create-vpc --cidr-block 10.0.0.0/16
{
  "Vpc": {
    "CidrBlock": "10.0.0.0/16",
    "DhcpOptionsId": "dopt-02824936f2eedfb9c",
    "State": "pending",
    "VpcId": "vpc-0c89c1e6752fc4dda",
    "OwnerId": "747621468838",
    "InstanceTenancy": "default",
    "Ipv6CidrBlockAssociationSet": [],
    "CidrBlockAssociationSet": [
      {
        "AssociationId": "vpc-cidr-assoc-0b348256d30997b2f",
        "CidrBlock": "10.0.0.0/16",
        "CidrBlockState": {
          "State": "associated"
        }
      }
    ]
  },
  "IsDefault": false
}
```

Create Subnets:- `aws ec2 create-subnet --vpc-id <your-vpc-id> --cidr-block 10.0.1.0/24`



```
PS C:\Users\Manasi> aws ec2 create-subnet --vpc-id vpc-0c89c1e6752fc4dda --cidr-block 10.0.1.0/24
{
  "Subnet": {
    "AvailabilityZone": "ap-south-1c",
    "AvailabilityZoneId": "aps1-az2",
    "AvailableIpAddressCount": 251,
    "CidrBlock": "10.0.1.0/24",
    "DefaultForAz": false,
    "MapPublicIpOnLaunch": false,
    "State": "available",
    "SubnetId": "subnet-0767304ed10585efe",
    "VpcId": "vpc-0c89c1e6752fc4dda",
    "OwnerId": "747621468838",
    "AssignIpv6AddressOnCreation": false,
    "Ipv6CidrBlockAssociationSet": [],
    "SubnetArn": "arn:aws:ec2:ap-south-1:747621468838:subnet/subnet-0767304ed10585efe",
    "EnableDns64": false,
    "Ipv6Native": false,
    "PrivateDnsNameOptionsOnLaunch": {
      "HostnameType": "ip-name",
      "EnableResourceNameDnsARecord": false,
      "EnableResourceNameDnsAAAARecord": false
    }
  }
}
```

`aws ec2 create-subnet --vpc-id <your-vpc-id> --cidr-block 10.0.2.0/24`



```
PS C:\Users\Manasi> aws ec2 create-subnet --vpc-id vpc-0c89c1e6752fc4dda --cidr-block 10.0.2.0/24
{
  "Subnet": {
    "AvailabilityZone": "ap-south-1c",
    "AvailabilityZoneId": "aps1-az2",
    "AvailableIpAddressCount": 251,
    "CidrBlock": "10.0.2.0/24",
    "DefaultForAz": false,
    "MapPublicIpOnLaunch": false,
    "State": "available",
    "SubnetId": "subnet-06e7cec47ee4d46b5",
    "VpcId": "vpc-0c89c1e6752fc4dda",
    "OwnerId": "747621468838",
    "AssignIpv6AddressOnCreation": false,
    "Ipv6CidrBlockAssociationSet": [],
    "SubnetArn": "arn:aws:ec2:ap-south-1:747621468838:subnet/subnet-06e7cec47ee4d46b5",
    "EnableDns64": false,
    "Ipv6Native": false,
    "PrivateDnsNameOptionsOnLaunch": {
      "HostnameType": "ip-name",
      "EnableResourceNameDnsARecord": false,
      "EnableResourceNameDnsAAAARecord": false
    }
  }
}
```

#Create Internet Gateway: `aws ec2 create-internet-gateway`

```
Windows PowerShell
PS C:\Users\Manasi> #Create Internet Gateway
PS C:\Users\Manasi> aws ec2 create-internet-gateway
{
  "InternetGateway": {
    "Attachments": [],
    "InternetGatewayId": "igw-058128abf01bb0f6b",
    "OwnerId": "747621468838",
    "Tags": []
  }
}
```

Attach Internet Gateway to VPC: - `aws ec2 attach-internet-gateway --vpc-id vpc-0c89c1e6752fc4dda --internet-gateway-id igw-058128abf01bb0f6b`

#Create Route Table : `aws ec2 create-route-table --vpc-id vpc-0c89c1e6752fc4dda`

```
Windows PowerShell
PS C:\Users\Manasi> aws ec2 attach-internet-gateway --vpc-id vpc-0c89c1e6752fc4dda --internet-gateway-id igw-058128abf01bb0f6b
PS C:\Users\Manasi> #Create Route Table
PS C:\Users\Manasi> aws ec2 create-route-table --vpc-id vpc-0c89c1e6752fc4dda
{
  "RouteTable": {
    "Associations": [],
    "PropagatingVgws": [],
    "RouteTableId": "rtb-0d672c2a476c654eb",
    "Routes": [
      {
        "DestinationCidrBlock": "10.0.0.0/16",
        "GatewayId": "local",
        "Origin": "CreateRouteTable",
        "State": "active"
      }
    ],
    "Tags": [],
    "VpcId": "vpc-0c89c1e6752fc4dda",
    "OwnerId": "747621468838"
  }
}
```

Create Route to Internet Gateway

P aws ec2 create-route --route-table-id rtb-0d672c2a476c654eb --destination-cidr-block 0.0.0.0/0 --gateway-id igw-058128abf01bb0f6b

```
Windows PowerShell
PS C:\Users\Manasi> #Create Route to Internet Gateway
PS C:\Users\Manasi> aws ec2 create-route --route-table-id rtb-0d672c2a476c654eb --destination-cidr-block 0.0.0.0/0 --gateway-id igw-058128abf01bb0f6b
{
  "Return": true
}
```

#Associate Subnets with Route Table:

aws ec2 associate-route-table --route-table-id rtb-0d672c2a476c654eb --subnet-id subnet-0767304ed10585efe

```
Windows PowerShell
PS C:\Users\Manasi> #Associate Subnets with Route Table
PS C:\Users\Manasi> aws ec2 associate-route-table --route-table-id rtb-0d672c2a476c654eb --subnet-id subnet-0767304ed10585efe
{
  "AssociationId": "rtbassoc-02669de768becab40",
  "AssociationState": {
    "State": "associated"
  }
}
```

aws ec2 associate-route-table --route-table-id rtb-0d672c2a476c654eb --subnet-id subnet-06e7cec47ee4d46b5

```
Windows PowerShell
PS C:\Users\Manasi> aws ec2 associate-route-table --route-table-id rtb-0d672c2a476c654eb --subnet-id subnet-06e7cec47ee4d46b5
{
  "AssociationId": "rtbassoc-04e5e5c465f411dff",
  "AssociationState": {
    "State": "associated"
  }
}
```

#Create Security Groups:- `aws ec2 create-security-group --vpc-id vpc-0c89c1e6752fc4dda --group-name bastion-sg --description "Bastion Security Group"`

```
PS C:\Users\Manasi> #Create Security Groups
PS C:\Users\Manasi> aws ec2 create-security-group --vpc-id vpc-0c89c1e6752fc4dda --group-name bastion-sg --description "Bastion Security Group"
{
  "GroupId": "sg-0db6ccd6cc998cf17"
}

PS C:\Users\Manasi> aws ec2 create-security-group --vpc-id vpc-0c89c1e6752fc4dda --group-name private-sg --description "Private Security Group"
{
  "GroupId": "sg-0d442483201b8f0fe"
}
```

#Allow Inbound SSH (Bastion) and Outbound (Private): `aws ec2 authorize-security-group-ingress --group-id sg-0db6ccd6cc998cf17 --protocol tcp --port 22 --cidr 0.0.0.0/0`

```
Windows PowerShell
PS C:\Users\Manasi> #Allow Inbound SSH (Bastion) and Outbound (Private)
PS C:\Users\Manasi> aws ec2 authorize-security-group-ingress --group-id sg-0db6ccd6cc998cf17 --protocol tcp --port 22 --cidr 0.0.0.0/0
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-09d28a183cc1bf173",
      "GroupId": "sg-0db6ccd6cc998cf17",
      "GroupOwnerId": "747621468838",
      "IsEgress": false,
      "IpProtocol": "tcp",
      "FromPort": 22,
      "ToPort": 22,
      "CidrIpv4": "0.0.0.0/0"
    }
  ]
}
```

```
aws ec2 authorize-security-group-egress --group-id sg-0d442483201b8f0fe --protocol tcp --port 22 --cidr 0.0.0.0/0
```

```
Windows PowerShell
PS C:\Users\Manasi> aws ec2 authorize-security-group-egress --group-id sg-0d442483201b8f0fe --protocol tcp --port 22 --cidr 0.0.0.0/0
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-079cf2a6dfe971ce1",
      "GroupId": "sg-0d442483201b8f0fe",
      "GroupOwnerId": "747621468838",
      "IsEgress": true,
      "IpProtocol": "tcp",
      "FromPort": 22,
      "ToPort": 22,
      "CidrIpv4": "0.0.0.0/0"
    }
  ]
}
```

```
#Launch Bastion and Private Instances: aws ec2 run-instances --image-id ami-026255a2746f88074 --subnet-id subnet-0767304ed10585efe --instance-type t3.small --security-group-ids sg-0db6ccd6cc998cf17 --key-name MyKeyPair
```

```
PS C:\Users\Manasi> aws ec2 run-instances --image-id ami-026255a2746f88074 --subnet-id subnet-0767304ed10585efe --instance-type t3.small --security-group-ids sg-0db6ccd6cc998cf17 --key-name MyKeyPair
{
  "Groups": [],
  "Instances": [
    {
      "AmiLaunchIndex": 0,
      "ImageId": "ami-026255a2746f88074",
      "InstanceId": "i-03782ff1e82ce0dfc",
      "InstanceType": "t3.small",
      "KeyName": "MyKeyPair",
      "LaunchTime": "2024-03-12T11:21:10+00:00",
      "Monitoring": {
        "State": "disabled"
      },
      "Placement": {
        "AvailabilityZone": "ap-south-1c",
        "GroupName": "",
        "Tenancy": "default"
      },
      "PrivateDnsName": "ip-10-0-1-34.ap-south-1.compute.internal",
      "PrivateIpAddress": "10.0.1.34",
      "ProductCodes": [],
      "PublicDnsName": "",
      "State": {
        "Code": 0,
        "Name": "pending"
      },
      "StateTransitionReason": "",
      "SubnetId": "subnet-0767304ed10585efe",
      "VpcId": "vpc-0c89c1e6752fc4dda",
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "ClientToken": "7f8af063-1036-41d9-b8fa-5a9942b3a6d1",
      "EbsOptimized": false,
      "EnaSupport": true,
      "Hypervisor": "xen",
      "NetworkInterfaces": [

```

```
{
  "Attachment": {
    "AttachTime": "2024-03-12T11:21:10+00:00",
    "AttachmentId": "eni-attach-0cfff45ee1c19a56",
    "DeleteOnTermination": true,
    "DeviceIndex": 0,
    "Status": "attaching",
    "NetworkCardIndex": 0
  },
  "Description": "",
  "Groups": [
    {
      "GroupName": "bastion-sg",
      "GroupId": "sg-0db6ccd6cc998cf17"
    }
  ],
  "Ipv6Addresses": [],
  "MacAddress": "06:89:ae:8b:67:43",
  "NetworkInterfaceId": "eni-0c07bcd7d232302c8",
  "OwnerId": "747621468838",
  "PrivateIpAddress": "10.0.1.34",
  "PrivateIpAddresses": [
    {
      "Primary": true,
      "PrivateIpAddress": "10.0.1.34"
    }
  ],
  "SourceDestCheck": true,
  "Status": "in-use",
  "SubnetId": "subnet-0767304ed10585efe",
  "VpcId": "vpc-0c89c1e6752fc4dda",
  "InterfaceType": "interface"
}
],
"RootDeviceName": "/dev/xvda",
"RootDeviceType": "ebs",
"SecurityGroups": [
  {
    "GroupName": "bastion-sg",
    "GroupId": "sg-0db6ccd6cc998cf17"
```

```
    },
    "SourceDestCheck": true,
    "StateReason": {
      "Code": "pending",
      "Message": "pending"
    },
    "VirtualizationType": "hvm",
    "CpuOptions": {
      "CoreCount": 1,
      "ThreadsPerCore": 2
    },
    "CapacityReservationSpecification": {
      "CapacityReservationPreference": "open"
    },
    "MetadataOptions": {
      "State": "pending",
      "HttpTokens": "optional",
      "HttpPutResponseHopLimit": 1,
      "HttpEndpoint": "enabled",
      "HttpProtocolIpv6": "disabled",
      "InstanceMetadataTags": "disabled"
    },
    "EnclaveOptions": {
      "Enabled": false
    },
    "PrivateDnsNameOptions": {
      "HostnameType": "ip-name",
      "EnableResourceNameDnsARecord": false,
      "EnableResourceNameDnsAAAARecord": false
    },
    "MaintenanceOptions": {
      "AutoRecovery": "default"
    },
    "CurrentInstanceBootMode": "legacy-bios"
  },
  "OwnerId": "747621468838",
  "ReservationId": "r-0ad89fa705ca78e88"
}
```

aws ec2 run-instances --image-id ami-026255a2746f88074 --subnet-id subnet-06e7cec47ee4d46b5 --instance-type t2.micro --security-group-ids sg-0d442483201b8f0fe --key-name vpc

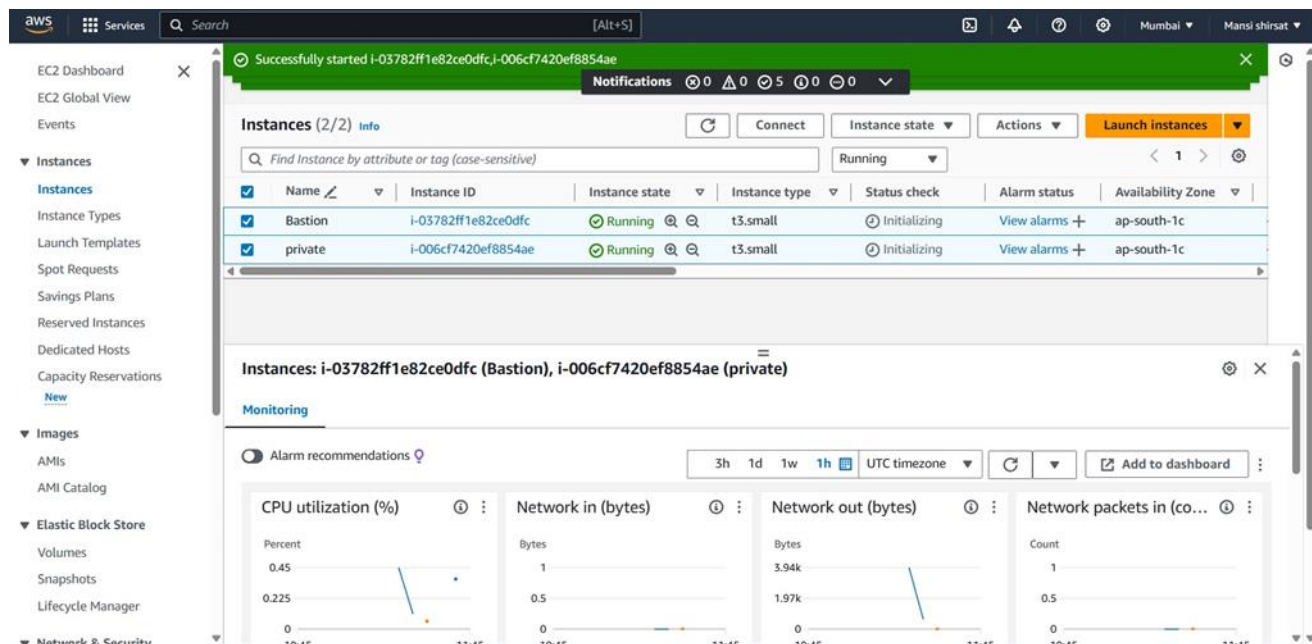
```
PS C:\Users\Manasi> aws ec2 run-instances --image-id ami-026255a2746f88074 --subnet-id subnet-06e7cec47ee4d46b5 --instance-type t3.small --security-group-id
sg-0d442483201b8f0fe --key-name vpc
{
  "Groups": [],
  "Instances": [
    {
      "AmiLaunchIndex": 0,
      "ImageId": "ami-026255a2746f88074",
      "InstanceId": "i-006cf7420ef8854ae",
      "InstanceType": "t3.small",
      "KeyName": "vpc",
      "LaunchTime": "2024-03-12T11:31:47+00:00",
      "Monitoring": {
        "State": "disabled"
      },
      "Placement": {
        "AvailabilityZone": "ap-south-1c",
        "GroupName": "",
        "Tenancy": "default"
      },
      "PrivateDnsName": "ip-10-0-2-199.ap-south-1.compute.internal",
      "PrivateIpAddress": "10.0.2.199",
      "ProductCodes": [],
      "PublicDnsName": "",
      "State": {
        "Code": 0,
        "Name": "pending"
      }
    }
  ]
}
```

```
Windows PowerShell X Windows PowerShell X + v
},
"StateTransitionReason": "",
"SubnetId": "subnet-06e7cec47ee4d46b5",
"VpcId": "vpc-0c89c1e6752fc4dda",
"Architecture": "x86_64",
"BlockDeviceMappings": [],
"ClientToken": "792963da-4aa1-4bb3-8663-726984fcbfab",
"EbsOptimized": false,
"EnaSupport": true,
"Hypervisor": "xen",
"NetworkInterfaces": [
  {
    "Attachment": {
      "AttachTime": "2024-03-12T11:31:47+00:00",
      "AttachmentId": "eni-attach-0b0daf299b1518ef5",
      "DeleteOnTermination": true,
      "DeviceIndex": 0,
      "Status": "attaching",
      "NetworkCardIndex": 0
    }
  }
],
```

```
,
"Description": "",
"Groups": [
  {
    "GroupName": "private-sg",
    "GroupId": "sg-0d442483201b8f0fe"
  }
],
"Ipv6Addresses": [],
"MacAddress": "06:62:6d:a0:b6:8b",
"NetworkInterfaceId": "eni-0e5d8e5cbc79b381a",
"OwnerId": "747621468838",
"PrivateIpAddress": "10.0.2.199",
"PrivateIpAddresses": [
  {
    "Primary": true,
    "PrivateIpAddress": "10.0.2.199"
  }
],
"SourceDestCheck": true,
"Status": "in-use",
```

```
Windows PowerShell
"CPUOptions": {
  "CoreCount": 1,
  "ThreadsPerCore": 2
},
"CapacityReservationSpecification": {
  "CapacityReservationPreference": "open"
},
"MetadataOptions": {
  "State": "pending",
  "HttpTokens": "optional",
  "HttpPutResponseHopLimit": 1,
  "HttpEndpoint": "enabled",
  "HttpProtocolIpv6": "disabled",
  "InstanceMetadataTags": "disabled"
},
"EnclaveOptions": {
  "Enabled": false
},
"PrivateDnsNameOptions": {
  "HostnameType": "ip-name",
  "EnableResourceNameDnsARecord": false,
  "EnableResourceNameDnsAAAARecord": false
},
"MaintenanceOptions": {
  "AutoRecovery": "default"
},
"CurrentInstanceBootMode": "legacy-bios"
},
"OwnerId": "747621468838",
"ReservationId": "r-03b3adff8e7531c56"
```

Output of above cli



Online Assessment x Security credentials | IAM | Glob... x Instances | EC2 | ap-south-1 x vpcs | VPC Console

https://ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#vpcs:

aws Services Search [Alt+S]

VPC dashboard x EC2 Global View EC2 Filter by VPC: Select a VPC Virtual private cloud Your VPCs

Your VPCs (1/2) Info

Search

Actions Create VPC

	Name	VPC ID	State	IPv4 CIDR	IPv6 CIDR
☐	-	vpc-0d26bae051ee633c7	Available	172.31.0.0/16	-
☒	Custom vpc	vpc-0c89c1e6752fc4dda	Available	10.0.0.0/16	-

Online Assessment x Security credentials | IAM | Glob... x Instances | EC2 | ap-south-1 x subnets | VPC Console

https://ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#subnets:

aws Services Search [Alt+S]

VPC dashboard x EC2 Global View EC2 Filter by VPC: Select a VPC Virtual private cloud Your VPCs Subnets Route tables Internet gateways Egress-only internet

Subnets (2/5) Info

Find resources by attribute or tag

Actions Create subnet

	Name	Subnet ID	State	VPC	IPv4 CIDR
☐	-	subnet-0c214d3c62bc0f092	Available	vpc-0d26bae051ee633c7	172.31.32.0/20
☐	-	subnet-044846ba4bd6868fa	Available	vpc-0d26bae051ee633c7	172.31.0.0/20
☒	Bastion	subnet-06e7cec47ee4d46b5	Available	vpc-0c89c1e6752fc4dda Cust...	10.0.2.0/24
☒	Private	subnet-0767304ed10585efe	Available	vpc-0c89c1e6752fc4dda Cust...	10.0.1.0/24
☐	-	subnet-04ca1052d8b2a5d12	Available	vpc-0d26bae051ee633c7	172.31.16.0/20

Online Assessment x Security credentials | IAM | Glob... x Instances | EC2 | ap-south-1 x RouteTables | VPC Console

https://ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#RouteTables:

aws Services Search [Alt+S]

VPC dashboard x EC2 Global View EC2 Filter by VPC: Select a VPC Virtual private cloud Your VPCs Subnets Route tables

Route tables (1/3) Info

Find resources by attribute or tag

Actions Create route table

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☒	Bastionrt	rtb-0d672c2a476c654eb	2 subnets	-	No	vpc-0c89c1e6752fc4dda
☐	-	rtb-0117cf37fbb4332a1	-	-	Yes	vpc-0c89c1e6752fc4dda
☐	-	rtb-0156e9e2eb23717ab	-	-	Yes	vpc-0d26bae051ee633c7

aws

Services

Search

[Alt+S]

Mumbai

Mansi shirsat

VPC dashboard

EC2 Global View

Filter by VPC:

Select a VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Internet gateways (1/1) Info

Search

Actions

Create internet gateway

	Name	Internet gateway ID	State	VPC ID	Owner
	-	igw-058128abf01bb0f6b	Attached	vpc-0c89c1e6752fc4dda Custom vpc	747621